Implementation and Evaluation of Brain Tumour Classification Using Convolutional Neural Networks

A REPORT

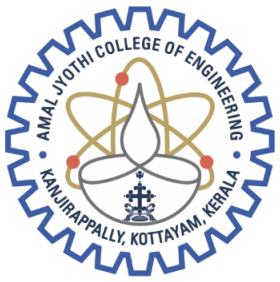
Submitted in partial fulfillment for the Case Study for

23ADT401 – Deep Learning

Department of IT & AI

Submitted by

Register No	Student Name
AJC23AD017	Ben Sebastian Joseph
AJC23AD021	Don Joseph
AJC23AD029	Joel Davis



AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

DEPARTMENT OF IT & AI

AMAL JYOTHI COLLEGE OF ENGINEERING

(AUTONOMOUS)

Kanjirappally, Kottayam, Kerala (INDIA)

October 2026

CONTENTS

No.	Title	Page No.
	Abstract	i
1	**Introduction**	**1**
1.1	Introduction	1
1.2	Background of Brain Tumour Classification	1
1.3	Deep Learning and Convolutional Neural Networks	1
1.4	Problem Statement	1
1.5	Objectives	1
1.6	Scope of the Study	2
2	**Literature Review**	**3**
2.1	Overview of the Selected Research	3
2.2	Architectures and Reported Results	3
2.3	Research Gap and Motivation	4
2.4	Summary	4
3	**Proposed Work**	**5**
3.1	Overview of the Proposed System	5
3.2	System Architecture	5
3.3	Dataset Description	5
3.4	Data Preprocessing	5
3.5	Duplicate Detection and Data Leakage Prevention	6
3.6	PaperGenericCNN Architecture	6
3.7	Training Workflow	6
3.8	Evaluation Methodology	7
3.9	Proposed Workflow	7
3.10	Summary	7
4	**Experimentation and Results**	**8**
4.1	Experimental Setup	8
4.2	Implementation Details	8
4.3	Training Configuration	8
4.4	Evaluation Metrics	8
4.5	Training Performance	8
4.6	Class-wise Performance Analysis	9
4.7	Comparison with the Research Paper	10
4.8	Limitations and Observations	11
5	**Conclusion and Scope for Further Research**	**12**
5.1	Conclusion	12
5.2	Limitations	12
5.3	Future Scope	12
	References	**13**

Abstract

Brain tumour classification from magnetic resonance imaging (MRI) is a multi-class image-recognition problem in which visual patterns can be subtle and vary across scans. This case study implements and evaluates a compact convolutional neural network (CNN), named *Paper-GenericCNN*, informed by the generic-CNN methodology in Gómez-Guzmán *et al.* [1]. The project inventory contains 7,200 images in four classes: 5,600 training images and 1,600 test images. The reference paper describes a separate, combined 7,023-image dataset [1]. The classes are glioma, meningioma, no tumour, and pituitary tumour. The training folder was stratified into 4,480 training and 1,120 validation images, with 280 validation images per class. A separate duplicate audit found 111 exact-duplicate groups and identified 114 duplicate test images. The 1,486 remaining clean test images were quarantined as well, so no image from the provided test folder was used for training, checkpoint selection, or evaluation.

The implemented network accepts RGB images resized to 256×256 pixels, scaled by $1/255$, and augmented during training with rotation, horizontal flipping, shear, zoom, and brightness transforms. It applies four convolutional blocks with channel counts 32, 64, 128, and 256, followed by a 65,536-feature flattening operation, dropout, and a four-class output layer. The model contains 651,044 parameters. It was trained on CPU for 10 epochs using Adam, a learning rate of 0.0001, batch size 16, and cross-entropy loss. The best validation checkpoint was Epoch 9: accuracy 80.98%, precision 85.33%, recall 80.98%, macro F1-score 81.86%, and loss 0.8291. Epoch 10 validation performance declined substantially, with the confusion matrix showing a marked increase in Meningioma predictions; the cause is unknown. These are validation results, not independent test results. They do not establish clinical utility or exact reproduction of the reference study; limited training, dataset differences, and the unavailable independent evaluation remain important limitations.

Keywords: Brain tumour classification, MRI, convolutional neural networks, data leakage, validation, deep learning.

Chapter 1
Introduction

1.1 Introduction

Magnetic resonance imaging provides non-invasive images of brain anatomy and is routinely important in the assessment of neurological conditions. Tumour classification from MRI is challenging because appearances vary in shape, location, contrast, and texture, while some visual patterns overlap between tumour categories. A convolutional neural network can learn image features from labelled examples and map those features to a predefined set of classes. In this case study, the task is limited to classifying an input image as glioma, meningioma, no tumour, or pituitary tumour. It is an educational machine-learning experiment, not a clinical diagnostic system.

1.2 Background of Brain Tumour Classification

The dataset uses three tumour labels and one non-tumour label. Gliomas, meningiomas, and pituitary tumours arise from different tissues and may show distinct imaging characteristics, but a single two-dimensional image does not capture the full clinical context. The class label supplied with an image is the target for supervised learning; it is not a segmentation mask and does not indicate tumour boundaries or provide a complete diagnosis. The dataset also combines images from multiple source collections, so acquisition and curation differences can become predictive shortcuts if they are correlated with class labels [1], [2].

1.3 Deep Learning and Convolutional Neural Networks

A CNN applies learned filters across local image regions. Early layers can respond to edges and simple textures, while deeper layers combine these responses into more abstract patterns. Weight sharing reduces the number of parameters compared with a fully connected layer applied directly to every input pixel. Pooling reduces spatial resolution and computation while making the representation less sensitive to small shifts. Batch normalization stabilizes intermediate activations during training, and dropout randomly suppresses activations during training to discourage reliance on a narrow set of features. These mechanisms support image classification but do not guarantee generalization to new scanners, institutions, or patient populations.

1.4 Problem Statement

The project asks whether a modest CNN trained on the supplied training folder can distinguish four MRI image classes under a controlled local setup. The main methodological concern is not only classification accuracy, but also whether evaluation data are kept separate from model fitting and whether duplicate images can produce leakage across dataset splits. The 1,600-image test folder was therefore excluded after an audit found exact duplicates; model selection and reported metrics use only the stratified validation subset formed from the training folder.

1.5 Objectives

The objectives are to implement a four-class CNN using the generic architecture family described in the selected paper; create a stratified training–validation split; audit exact duplicates and prevent test-folder exposure; measure validation performance using aggregate and class-wise metrics; and compare results with the paper while clearly identifying differences in data, architecture, training, and evaluation. A further objective is to state limitations without interpreting benchmark performance as evidence of clinical readiness.

1.6 Scope of the Study

The work covers image-level classification of the four supplied labels using one custom CNN and one validation split. It reports the given model configuration, documented resizing, scaling and training augmentation, CPU environment, run duration, memory footprint, and validation scores. It does not evaluate transfer-learning models, segmentation, localization, prospective clinical data, or the quarantined test images. No patient-level grouping procedure is assumed. The conclusions apply only to this experiment and its validation partition.

Chapter 2
Literature Review

2.1 Overview of the Selected Research

The primary reference is the study by Gómez-Guzmán *et al.*, which compares a generic CNN with six pre-trained CNN architectures for four-class brain MRI classification [1]. The article describes a dataset assembled from Figshare, SARTAJ, and Br35H sources and reports 7,023 images. The authors consider both a generic network and transfer-learning models, use preprocessing and augmentation, and describe a k-fold evaluation protocol; the exact fold count is not asserted here. Its highest mean accuracy is 97.12% for InceptionV3. Those figures describe the paper's protocol and are not results of the present implementation.

2.2 Architectures and Reported Results

The paper's model set represents different feature-extraction strategies. The generic CNN learns all convolutional filters from the study data. ResNet50 uses residual connections that help information and gradients pass through a deep network. InceptionV3 combines multi-scale convolutional branches and factorized operations. InceptionResNetV2 adds residual connections to Inception-style modules. Xception relies on depthwise-separable convolutions to separate spatial and channel mixing. MobileNetV2 uses efficient depthwise operations and inverted residual bottlenecks, targeting lower computation. EfficientNetB0 applies balanced compound scaling of network depth, width, and input resolution. In the cited comparison these are evaluated as transfer-learning alternatives; their pretrained weights and heads are not part of our custom CNN.

Table 2.1: Accuracy of CNN Architectures Reported in [1]

Model	Main architectural idea	Accuracy
Generic CNN	Four convolution stages; features learned for the task	81.08%
ResNet50	Residual skip connections	96.97%
InceptionV3	Parallel multi-scale Inception modules	97.12%
InceptionResNetV2	Inception modules with residual connections	96.78%
Xception	Depthwise-separable convolution	95.67%
MobileNetV2	Inverted residual and depthwise blocks	95.45%
EfficientNetB0	Compound depth, width, and resolution scaling	90.88%

In the paper's performance table, InceptionV3, ResNet50, and InceptionResNetV2 are the three highest-accuracy models, followed by Xception and MobileNetV2; EfficientNetB0 and the

generic CNN score lower on that protocol [1]. The article's results narrative gives 81.05% for the generic CNN, whereas its comparison table lists 81.08%. The table value is retained above and in the later comparison to match the tabulated metrics; the small internal discrepancy is noted rather than silently reconciled. The reference also reports 97.12% as the five-fold mean for InceptionV3. These are paper-reported outcomes on its own dataset and experimental setup.

2.3 Research Gap and Motivation

A high score from a transfer-learning model does not establish that a smaller CNN trained from scratch will perform similarly, nor does a random image split ensure independence when duplicate or related slices are present. The present work focuses on the generic-CNN direction and emphasizes data partition hygiene, a transparent parameter count, and validation-only reporting. Its architecture is informed by the reference's generic CNN, but its exact filter widths and training configuration are implementation choices. The experiment therefore offers a bounded comparison rather than a replication of all seven networks or the paper's five-fold protocol.

2.4 Summary

The reviewed study compares a task-trained generic CNN with six transfer-learning architectures and reports its strongest result for InceptionV3. These findings motivate a careful implementation of the generic-CNN approach, while differences in dataset handling and training prevent a claim of exact reproduction.

Chapter 3
Proposed Work

3.1 Overview of the Proposed System

The proposed system maps one resized RGB MRI image to one of four class scores: glioma, meningioma, no tumour, or pituitary tumour. It is a supervised image-classification workflow informed by the generic-CNN direction of the selected research paper. The project evaluates one custom CNN and reports validation performance; it does not perform segmentation, tumour localization, or clinical diagnosis.

3.2 System Architecture

The system has four stages: image curation and split control, image preprocessing, feature extraction and classification by *PaperGenericCNN*, and validation-based checkpoint assessment. Duplicate auditing and quarantine isolate the test directory, while validation is used only for checkpoint selection and reported metrics. The project records do not establish patient-level grouping, so image-level stratification should not be interpreted as patient-independent validation.

3.3 Dataset Description

The project inventory is the Kaggle "Brain Tumor MRI Dataset" by Masoud Nickparvar [2]; its exact source URL is:
`https://www.kaggle.com/`
`datasets/masoudnickparvar/brain-tumor-mri-dataset`
The inventory contains 7,200 images: 5,600 in the training directory and 1,600 in the testing directory. The 7,023-image total belongs to the separate combined dataset described in the reference paper [1], not the count used for this project inventory. Four directory labels are used: glioma, meningioma, no tumour, and pituitary. The training directory was deterministically divided into 4,480 training and 1,120 validation images, with 280 validation images per class. The duplicate audit identified 114 test duplicates; the remaining 1,486 test images were quarantined and neither subset was used in fitting or metric calculation.

3.4 Data Preprocessing

Inputs are represented as three-channel RGB images, resized to 256×256 pixels, and scaled by $1/255$. Training-only augmentation uses rotation of 10°, horizontal flipping with probability 0.5, shear of 10°, zoom in the range $[0.9, 1.1]$, and brightness in the configured range $[0, 0.7]$. The supplied presentation does not specify interpolation details or the exact implementation convention for the brightness range; these are not inferred. Validation images are not augmented. Transform choices can affect learned features and validation results, so they should be recorded when reproducing the experiment.

3.5 Duplicate Detection and Data Leakage Prevention

An exact-duplicate audit identified 111 duplicate groups across the supplied training and testing folders and found 114 test images that matched duplicates. Because a test image appearing in training can inflate apparent performance, all test-folder images were quarantined from model development, including the 1,486 images not flagged as duplicates. They were not loaded for evaluation and did not inform model selection. This is a conservative precaution, not evidence that the remaining images are independent; exact-duplicate screening does not rule out near-duplicates, multiple images from the same patient, or source-level correlations. The resulting validation score still is not a final unbiased estimate because validation was used to select the best epoch and may share source-level or patient-level correlations with training data.

3.6 PaperGenericCNN Architecture

The implemented model, *PaperGenericCNN*, has four repeated blocks. Each block applies a 3×3 convolution, batch normalization, ReLU, and 2×2 max pooling. The channel progression is $3 \to 32 \to 64 \to 128 \to 256$, while the spatial progression is $256 \to 128 \to 64 \to 32 \to 16$. Thus, the final feature map is $256 \times 16 \times 16$, or 65,536 flattened features. A dropout layer with probability $p = 0.5$ precedes a fully connected layer producing four class logits. The project-reported total is 651,044 parameters. The four filter widths are specific to this implementation; they are not asserted as confirmed details of every configuration in the original paper.

For an input tensor X, convolution forms feature maps by applying learned kernels across local neighborhoods; in simplified notation,

$$Y_{i,j,k} = \sum_{u,v,c} W_{u,v,c,k} X_{i+u,j+v,c} + b_k.$$

Batch normalization rescales activations using batch statistics and learned scale and shift parameters. ReLU applies $\max(0, x)$ element-wise, and max pooling retains the largest activation in each local window. The successive reductions lower spatial cost while increasing the number of learned channels. Flattening then presents the learned features to the classifier. Dropout is active during training to regularize the dense decision stage; it is disabled for validation inference. The final logits are converted to class probabilities by softmax when probabilities are needed.

3.7 Training Workflow

The training partition is passed in mini-batches of 16. For each batch, the model computes four logits per image, evaluates PyTorch CrossEntropyLoss against the integer class labels, backpropagates gradients, and updates parameters with Adam at learning rate 10^{-4} ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The reported E2 setup uses no label smoothing or weight decay. The process runs for 10 epochs, with validation metrics used to monitor performance and retain the best checkpoint. Validation examples are not part of gradient updates. Epoch 9 was selected as

the best observed validation checkpoint; no test-folder metric informed selection. No early-stopping patience, scheduler, random-seed value, or checkpoint-selection code is available in the supplied information; such details are not inferred.

3.8 Evaluation Methodology

Evaluation uses only the 1,120-image validation split. With TP, TN, FP, and FN denoting the relevant class-wise counts, accuracy is the fraction of correct predictions; precision is $TP/(TP + FP)$, recall is $TP/(TP + FN)$, and the F1-score is the harmonic mean of precision and recall. The reported class-wise values are summarized across four classes, and the macro F1 is the unweighted mean of per-class F1 scores. Cross-entropy loss measures the negative log probability assigned to the correct labels. The untouched test folder is not used to report any metric.

3.9 Proposed Workflow

The implementation follows these steps:

1. Audit the 5,600-image training folder and 1,600-image test folder for exact duplicates; keep all test-folder images quarantined.
2. Create the deterministic stratified split of 4,480 training and 1,120 validation images, with 280 validation images per class.
3. Convert inputs to RGB, resize and scale them, and apply the documented augmentations to training images only.
4. Train *PaperGenericCNN* with the stated E2 configuration for 10 epochs.
5. Select the best checkpoint using validation performance and report aggregate and class-wise validation metrics; do not evaluate the quarantined test images.

3.10 Summary

The proposed work evaluates one compact, task-trained CNN under a leakage-aware split. Its reported outcomes are validation metrics only, with the independent test folder left unevaluated.

Chapter 4
Experimentation and Results

4.1 Experimental Setup

The experiment was conducted on Arch Linux, x86_64, kernel 7.1.5, with an Intel Core i7-1255U CPU and 7.4 GiB RAM. Training used CPU-only execution, Python 3.12.7, and PyTorch 2.14.1+cpu as recorded in the project materials. These details contextualize this run; elapsed times should not be compared across papers as a hardware-independent measure.

4.2 Implementation Details

The model was implemented in PyTorch as *PaperGenericCNN*. Input files were converted to RGB, resized to 256×256 pixels, and scaled by $1/255$. Training-only augmentation used rotation up to $10°$, horizontal flip probability 0.5, shear up to $10°$, zoom range $[0.9, 1.1]$, and brightness range $[0, 0.7]$. The deterministic stratified split contains 4,480 training and 1,120 validation images, with equal class support in validation. The supplied presentation does not specify image interpolation or the brightness transform's implementation convention.

4.3 Training Configuration

Training used batch size 16, Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with learning rate 0.0001, CrossEntropyLoss without label smoothing or weight decay, dropout probability 0.5, and 10 epochs. Total training time was approximately 61.4 minutes and peak memory footprint approximately 606 MB. Checkpoint selection used validation performance; Epoch 9 was retained as the best checkpoint. No random seed, scheduler, or stopping-patience value is available in the supplied information.

4.4 Evaluation Metrics

Accuracy measures the proportion of all validation examples assigned the correct class. Precision measures how often predictions for a class are correct, while recall measures how many examples of that class are recovered. The F1-score combines precision and recall, and macro F1 gives each class equal weight. On this balanced validation split, per-class support is 280, which makes macro recall numerically equal to overall accuracy for a single-label, four-class task. Cross-entropy loss is the mean negative log probability assigned to correct labels; its scale depends on confidence and implementation details.

4.5 Training Performance

The complete history reported for E2 is shown below. Epoch 9 was the best validation checkpoint: it achieved 80.98% accuracy, 85.33% macro precision, 80.98% macro recall, 81.86% macro F1, and validation loss 0.8291. At Epoch 10, training accuracy continued to rise, but validation accuracy fell to 70.71%, macro F1 to 70.99%, and loss rose to 1.8991.

Table 4.1: E2 Training and Validation History

Epoch	Train loss	Train acc.	Train macro-F1	Val. loss	Val. acc.	Val. macro-F1
1	1.4350	53.26%	0.5341	1.4650	67.23%	0.6742
2	1.1114	62.75%	0.6280	1.0994	72.23%	0.7127
3	0.9460	66.12%	0.6613	1.3590	65.89%	0.6562
4	0.8974	68.95%	0.6899	1.2429	71.43%	0.7019
5	0.8524	70.07%	0.7007	1.0941	72.50%	0.7084
6	0.8216	73.37%	0.7343	0.8767	76.34%	0.7559
7	0.8206	72.52%	0.7251	1.3858	73.39%	0.7269
8	0.6866	75.87%	0.7589	1.0374	78.39%	0.7878
9	0.6520	76.72%	0.7674	0.8291	80.98%	0.8186
10	0.6701	77.34%	0.7733	1.8991	70.71%	0.7099

The Epoch 10 confusion matrix contains 584 predictions as Meningioma, compared with 417 at Epoch 9. This coincides with the validation decline, but the cause is unknown; brightness dimming is only a proposed hypothesis and has not been isolated experimentally. The selected checkpoint is the best observed validation epoch, not an independent test result.

4.6 Class-wise Performance Analysis

The per-class validation results reveal different error patterns:

Table 4.2: Epoch 9 Class-wise Validation Metrics

Class	Precision	Recall	F1-score
Glioma	92.37%	77.86%	84.50%
Meningioma	59.23%	88.21%	70.88%
No Tumor	97.69%	75.36%	85.08%
Pituitary	92.03%	82.50%	87.01%

The Epoch 9 validation confusion matrix gives the counts underlying these patterns:

Table 4.3: Epoch 9 Validation Confusion Matrix

Actual \ Predicted	Glioma	Meningioma	No Tumor	Pituitary	Total
Glioma	218	58	0	4	280
Meningioma	14	247	5	14	280
No Tumor	3	64	211	2	280
Pituitary	1	48	0	231	280
Predicted total	236	417	216	251	1,120

Meningioma recall is relatively high, but its precision is substantially lower: 170 non-meningioma images were predicted as Meningioma. No Tumor has the highest precision but lower recall, consistent with missed no-tumour examples. Glioma precision is high but recall is lower, while pituitary performance is comparatively balanced. The matrix quantifies these validation errors but does not identify their visual causes; training-source example filenames in the presentation were not verified as validation predictions.

4.7 Comparison with the Research Paper

The following comparison places the project's Epoch 9 validation results beside the generic-CNN figures in the research-paper table [1].

Table 4.4: Paper Results and E2 Validation Comparison

Metric	Paper Generic CNN	Our E2 validation	Paper InceptionV3
Accuracy	81.08%	80.98%	97.12%
Precision	85.27%	85.33%	97.97%
Recall	76.77%	80.98%	96.59%
Macro F1-score	Not reported	81.86%	Not reported
Cross-entropy loss	0.4661	0.8291	0.0796
Parameters	8.7 million	651,044	23.9 million
Training time	200 min	61.4 min	323 min

E2 accuracy is 0.10 percentage points below the paper's Generic CNN; this similarity is descriptive and does not establish exact reproduction. The paper reports model results under its own protocol, including cross-validation, while this project uses one deterministic stratified

validation split from its local 7,200-image inventory and a quarantined test folder. InceptionV3 is a separately pretrained model with 23.9 million parameters, so its stronger reported results are not a like-for-like comparison with E2. The paper's Generic CNN configuration is internally inconsistent: Table 2 identifies Adam, whereas the text identifies SGD; the table gives dropout 0.5, while the text gives 0.2. E2 uses Adam and dropout 0.5, but the exact paper configuration cannot be established. E2's filter counts are implementation choices, and its parameter count is much smaller. Different partitions, preprocessing, training recipes, and aggregation conventions can change metrics. The paper table reports generic-CNN accuracy as 81.08%, although its narrative states 81.05%; the table value is retained here. Cross-entropy values and training times are hardware- and protocol-dependent. Most importantly, 80.98% is validation performance only, not an independent test score.

4.8 Limitations and Observations

The experiment is based on one 10-epoch run and one deterministic stratified validation split. The presentation reports the preprocessing ranges, complete epoch history, and Epoch 9 and Epoch 10 confusion matrices, but does not provide a random seed, patient identifiers, image-interpolation details, or source-level grouping. Exact-duplicate screening cannot rule out near-duplicates, adjacent slices, or images from the same patient crossing splits. The 1,486 clean test images were not used, so no independent final estimate is available. Results from an image-level dataset assembled across sources may encode source-specific artifacts. The model predicts dataset labels only; no external cohort, prospective study, calibration analysis, or clinical workflow evaluation was performed.

Chapter 5
Conclusion and Scope for Further Research

5.1 Conclusion

This case study implemented a four-class CNN with 651,044 parameters and evaluated it on a stratified validation subset drawn solely from the training folder. The Epoch 9 checkpoint achieved 80.98% accuracy and 81.86% macro F1 on 1,120 validation images. Per-class results show that performance is not uniform, particularly the low meningioma precision and reduced recall for the no-tumour class. The separate test directory was withheld after duplicate auditing and was not evaluated. Consequently, the study demonstrates an implementation workflow and validation result, not an independent generalization estimate. Differences in dataset version and training protocol also prevent a claim of exact replication of Gómez-Guzmán *et al.* [1].

5.2 Limitations

The study uses one short run and stratified split; the random seed and some preprocessing details are unrecorded, and patient-level grouping is unknown. Exact-duplicate screening cannot exclude related slices or source-level correlations. Validation informed checkpoint selection, and no independent test, external-cohort, calibration, prospective, or clinical-workflow evaluation was performed. The model is not a substitute for professional assessment.

5.3 Future Scope

Future work should document seeds, preprocessing, checkpoint rules, and software versions, then repeat runs across controlled seeds. Ablation studies should test augmentation, including the proposed E3 brightness-jitter range $[0.7, 1.3]$, which has not been tested. Compare class-weighted loss or balanced sampling for meningioma precision and transfer-learning baselines under common splits and compute budgets. After model selection, evaluate an independent clean or external test set once. Clinical use would require representative multi-site data, expert oversight, prospective validation, calibration, safety, and regulatory review; saliency maps alone do not establish validity.

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